

U12A2MA  
Multimedia and Animation

Unit-2 : Introduction to Image Editing (Using  
Photoshop)

## Q-1. Explain Transformation in Photoshop.

### Transform submenu commands

#### Scale

Enlarges or reduces an item relative to its reference point, the fixed point around which transformations are performed. You can scale horizontally, vertically, or both horizontally and vertically.

#### Rotate

Turns an item around a reference point. By default, this point is at the center of the object; however, you can move it to another location.

#### Skew

Slants (to give slope) an item vertically and horizontally.

#### Distort

Stretches an item in all directions. To twist image in all ways.

#### Perspective

Applies one-point perspective to an item.

#### Warp

Manipulates the shape of an item.

#### Rotate 180, Rotate 90 CW, Rotate 90 CCW

Rotates the item by the specified number of degrees, either clockwise or counterclockwise.

#### Flip

Flips the item vertically or horizontally.

## eQ-2. Write a short note on HSB (Hue, Saturation and Brightness) or HSL.

**Hue** — *another word for color*

**Saturation**— *the intensity or purity of a hue*

**Lightness** — *the relative degree of black or white mixed with a given hue*

Hue, saturation, and brightness are features of color in the red, green, and blue ( RGB ) scheme. These terms are most often used in reference to the color of each pixel. All possible colors can be specified according to hue, saturation, and brightness (also

called brilliance), just as colors can be represented in terms of the R, G, and B components.

Hue is therefore the actual color. Whatever we can see is dependent on the wavelength of light being reflected or produced. So, hue is wavelength in within light.

Saturation indicates the amount of grey in a color. 100% saturation means there's no addition of gray to the hue. The color is completely pure. At the other extreme a hue with 0% saturation appears as a medium gray. So that as saturation increases, colors appear more "pure" and as saturation decreases, colors appear more "washed-out" (like fade up).

Brightness refers to how much white (or black) is mixed in the color (means mixed with a given hue). Adding white makes the color lighter (creates tints) and adding black makes it darker (creates shades).

### Q-3. Explain various options of paint brush in photoshop.

**Shape Dynamics:** Shape Dynamics option let you adjust size of brush, angel of brush, spacing between stroke, and roundness to the brush.

**Scattering:**Brush scattering determines the number and placement of marks in a stroke. Specifies how brush marks are distributed in a stroke.

**Texture:**options allow you to use texture in brush stroke.

**Dual Brush:** option let you use multiple brushes in single stroke. It is something like brush within brush.

**Color Dynamics:** Allow to work with brushing color. We can mix foreground and background color, adjust hue, saturation and purity of color also.

**Transfer:**to provide transparent effect to brush stroke. We can also adjust opacity of brush here.

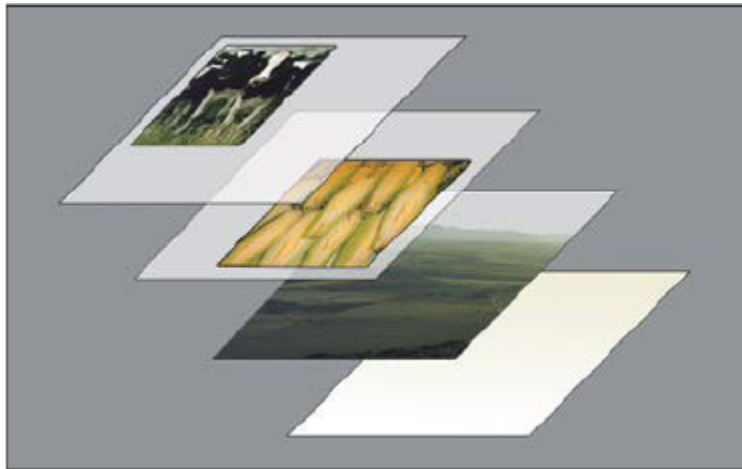
**Brush Pose:**Brush pose options let you achieve stylus-like effects and let you control the angle and position of the brush.

**Noise:**Add some noise to soft edge.

**Wet Edge:** Causes paint to build up along the edges of the brush stroke, creating a watercolor effect.

#### Q-4. Write a short-note on layer and layer mask.

Photoshop layers are like sheets of stacked acetate. You can see through transparent areas of a layer to the layers below. You move a layer to position the content on the layer, like sliding a sheet of acetate in a stack. You can also change the opacity of a layer to make content partially transparent.



*Transparent areas on a layer let you see layers below.*

You use layers to perform tasks such as compositing multiple images, adding text to an image, or adding vector graphic shapes. You can apply a layer style to add a special effect such as a drop shadow or a glow.

#### **Layer Mask**

Layer masks are resolution-dependent bitmap images that are edited with the painting or selection tools.

You can add a mask to a layer and use the mask to hide portions of the layer and expose the layers below. Layer and vector masks are nondestructive, which means you can go back and re-edit the masks later without losing the pixels they hide.

### Q-5. Elaborate the Blur filters in brief.

The Blur filters soften a selection or an entire image, and are useful for retouching. They smooth transitions by averaging the pixels next to the hard edges of defined lines and shaded areas in an image.



#### Average

Finds the average color of an image or selection, and then fills the image or selection with the color to create a smooth look. For example, if you select an area of grass, the filter changes the area into a homogeneous patch of green.

#### Blur and Blur More

Eliminate noise where significant color transitions occur in an image. Blur filters smooth transitions by averaging the pixels next to the hard edges of defined lines and shaded areas. The effect of the Blur More filter is three or four times stronger than that of the Blur filter.

#### Box Blur

Blurs an image based on the average color value of neighboring pixels. This filter is useful for creating special effects. You can adjust the size of the area used to calculate the average value for a given pixel; a larger radius results in greater blurring.

#### Gaussian Blur

Quickly blurs a selection by an adjustable amount. *Gaussian* refers to the bell-shaped curve that is generated when Photoshop applies a weighted average to the pixels. The Gaussian Blur filter adds low-frequency detail and can produce a hazy effect.

## **Lens Blur**

Adds blur to an image to give the effect of a narrower depth of field so that some objects in the image stay in focus and others areas are blurred. See Add lens blur.

## **Radial Blur**

Simulates the blur of a zooming or rotating camera to produce a soft blur. Choose Spin to blur along concentric circular lines, and then specify a degree of rotation. Choose Zoom to blur along radial lines, as if zooming into or out of the image, and specify a value from 1 to 100. Blur quality ranges from Draft (for fast but grainy results) or Good and Best for smoother results, which are indistinguishable from each other except on a large selection. Specify the origin of the blur by dragging the pattern in the Blur Center box.

## **Shape Blur**

Uses the specified kernel to create the blur. Choose a kernel from the list of custom shape presets, and use the radius slider to adjust its size. You can load different shape libraries by clicking the triangle and choosing from the list. Radius determines the size of the kernel; the larger the kernel, the greater the blur.

## **Smart Blur**

Blurs an image with precision. You can specify a radius, a threshold, and a blur quality. The Radius value determines the size of the area searched for dissimilar pixels. The Threshold value determines how dissimilar the pixels must be before they are affected. You also can set a mode for the entire selection (Normal) or for the edges of color transitions (Edge Only and Overlay Edge). Where significant contrast occurs, Edge Only applies black-and-white edges, and Overlay Edge applies white.

## **Surface Blur**

Blurs an image while preserving edges. This filter is useful for creating special effects and for removing noise or graininess. The Radius option specifies the size of the area sampled for the blur. The Threshold option controls how much the tonal values of neighboring pixels must diverge from the center pixel value before being part of the blur. Pixels with tonal value differences less than the Threshold value are excluded from the blur.

## **Q-6. Answer the followings.**

### **What is a Gaussian blur?**

Gaussian blur is a tool that blurs the edges of the image to make it more real and adaptive to the surrounding. Often it is believed that Gaussian blur can only blur the image, but if it is used cleverly it will not blur the image but enhance the appearance.

### **How does Red Eye tool works?**

Red eye tool is a type of a repairing tool. When you circle any problematic area with red eye tool, the Photoshop will automatically identify that area and will replace it by dark-gray pixels.

### **Explain about the Photoshop Work Area?**

Ans:- Photoshop work area consists of Application Bar, Tools Panel, Options Bar, Document Window and Panel Dock.

- Application Bar: This contains menu bar along with various controls to zoom plus and minus, buttons.
- Tools Panel: This tool has a set of tools for selecting, manipulating images, artwork etc.
- Options Bar: There are various options for currently selected tools.
- Document Window: It shows the current working file. It can be grouped and docked.
- Panel Dock: Panel dock has several panels to modify the images. By default, it has a layer, adjustment, and style panels.